

TA Guide — Lab 5: pavo Introduction

Quantifying reef-fish color patterns (Week 6)

EEB 187 — Ecology & Evolution of Color

May 6, 2026

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Purpose of this guide

This document is for **Rosamari** (and any future TA stepping into Lab 5). It pairs with four other things:

Artifact	Where	Purpose
<code>scripts/walkthrough.R</code>	repo root	the live-coded R script you and the class will run together
<code>scripts/helpers.R</code>	repo root	helper functions students source at session start
<code>lab-05-pavo-intro.qmd</code>	repo root	the student worksheet with Part A (demo) + Part B (their three fish)
<code>student-guide.qmd</code>	repo root	self-paced step-by-step guide students follow during/after lab
<code>pre-lab-installation.qmd</code>	repo root	what students were asked to install before lab

i Recommended student launch path: Docker

Students were told (in the BruinLearn announcement and on the prep page) to use **Docker** as their primary path. The `ghcr.io/alfarolab/eeb187-pavo-lab:latest` image is pre-baked and verified end-to-end. Other paths (Native, Binder, Codespaces) work too — Binder has been independently verified after this morning’s fixes — but Docker is the recipe everyone should default to.

The other paths are kept around as fallbacks; if a student says “*Docker isn’t working*,” recommend they try Binder (<https://mybinder.org/v2/gh/AlfaroLab/eeb187-pavo-lab/HEAD?urlpath=rstudio>) — it’s a 2-minute fallback that requires zero install.

This guide goes **deeper than the slide deck** on the conceptual basis for everything in `walkthrough.R`, then walks through the script step by step with **(a) what it computes, (b) why it matters biologically, (c) student questions to expect, and (d) troubleshooting for the errors you will actually see in lab.**

Read this *before* the first time you teach the lab. Skim it again the morning of, especially the troubleshooting boxes. Total reading time: ~45 minutes.

Part 1 — What the lab teaches

The pedagogical claim

Color *patterns* — vertical bars, horizontal stripes, reticulate spots — can be reduced to a small set of numbers that distinguish them automatically. The single most important number is the **A statistic** (aspect ratio).

By the end of lab the students should be able to:

1. Predict the rough A value of a fish from looking at its photo ($A > 1$, $A < 1$, $A \approx 1$).
2. Run the pipeline themselves on a new fish image (segment $\rightarrow$ classify $\rightarrow$ adjacent $\rightarrow$ metrics).
3. Explain to a peer *why* A distinguishes vertical from horizontal patterns — at the level of “we count adjacent pixel pairs and direction matters.”
4. Critique a result. When they get unexpected metrics, they should be able to point to *which step* (segmentation, k choice, white-background exclusion) is the likely culprit.

What’s NOT in scope today

These come later in the term:

- Spectral analysis from spectrometers (`procspec`, `peakshape`)
- Visual modeling through animal eyes (`vismodel`, `coldist`, JNDs) — Lab 6
- Many-species comparative analysis with phylogeny — Lab 9
- Endler’s saturation/luminance space arguments at the *receiver* level — Lecture 14–16

If a student asks “but what does the *fish that’s looking at this fish* see?” — that’s the right question, and the answer is “we’ll get there in Lab 6 with `vismodel()`.” For today, every metric is **photographic**, not perceptual.

Part 2 — Conceptual primer

This is the layer below the slide deck — the math and the *why*. Students don’t need all of this. You do.

2.1 The Endler 2012 framework

The whole pipeline is built on John Endler’s 2012 *Biol. J. Linn. Soc.* paper “A framework for analysing colour pattern geometry: adjacent colours.” The core move:

1. **Discretize** the image into a small number of color classes (k-means in RGB).
2. **Sample** the classified image on a regular grid.
3. **Count** transitions between color classes between adjacent grid cells — separately for horizontal vs. vertical adjacency.
4. **Summarize** the transition matrix with a small set of geometry-aware statistics.

Step 4 produces the metrics students will see:

Metric	What it measures	Intuition
k	number of color classes	how many “colors” the fish has
m	transition density (off-diagonal mass / total)	how <i>busy</i> the pattern is per unit area
A	vertical edges / horizontal edges (the aspect ratio)	direction — bars vs. stripes vs. isotropic
Jc	color heterogeneity (Simpson on color-class areas)	how <i>evenly</i> the colors are used
Jt	transition heterogeneity (Simpson on transitions)	how diverse the <i>boundaries</i> are
m_dS	mean chromatic boundary strength	how <i>saturated</i> a typical edge is
m_dL	mean luminance boundary strength	how <i>bright/dark</i> a typical edge is

The A statistic is the headline because:

- It cleanly distinguishes **bars** ($A > 1$), **stripes** ($A < 1$), and **isotropic patterns** ($A \approx 1$) in three lines of R.
- It’s invariant to color choice — a yellow-and-black bar code and a red-and-white bar code give the same A.
- It’s a *geometric* claim about pattern, not a perceptual one — so it’s a clean entry point before we add receiver biology in Lab 6.

2.2 Why segmentation matters (Step 1.5 of walkthrough)

The full pipeline is `image` → `classify` → `adjacent` → `metrics`. Every step downstream of `classify()` is poisoned by background pixels:

- **k-means** allocates color classes to *whatever pixels exist*. If half the photo is blue water, one of your k classes will be water, and you’ll have one less class for the fish.
- **Adjacent grid sampling** drops a regular $xpts \times ypts$ grid over the *whole image*. Long uniform stretches of background drown out the fish’s directional signal and pull A toward 1.

Three demo fish on uncropped photos will give $A = 1$ across the board — the lab’s punchline disappears. **The dramatic A values (1.54 / 0.60 / 1.02) are entirely a product of segmentation.** This is why the demo fish are pre-segmented and why Step 1.5 has three Paths.

The white-background trick

We segment to a white background and then *include* white as a +1 in the `k` argument: `kcols = (fish_colors + 1)`. The white class becomes one of the `k` centroids, and `bkgID + exclude = "background"` tells `adjacent()` to ignore transitions involving white. Net effect: same as analyzing only the fish, but in one R call without a polygon mask.

2.3 What `classify()` actually does

`classify(img, kcols = k)` runs **k-means on each pixel’s RGB triplet**:

1. Treats each pixel as a point in 3D color space (R, G, B).
2. Initializes `k` centroids randomly.
3. Iteratively assigns each pixel to its nearest centroid, then moves each centroid to the centroid of its assigned pixels. Repeats until stable.
4. Returns an integer-valued array (the “classified image”), plus a `classRGB` attribute giving the `k` centroid colors.

This is a *purely color-based* clustering — it knows nothing about spatial structure. A 3-pixel red dot in the corner gets the same class as a 3-pixel red dot on the body. Spatial structure enters at the *next* step.

k-means is non-deterministic without a seed

k-means initializes centroids randomly. Two runs on the same image with the same `k` can produce slightly different palettes. pavo’s `classify()` uses a fixed seed under the hood for reproducibility, but if students compare each other’s runs, the *order* of color classes (class 1, 2, 3, ...) may differ even when the colors themselves are the same.

Practical implication: when comparing two students’ fish, compare RGB centroids in `classRGB`, not class indices.

2.4 What `adjacent()` actually does

`adjacent()` is the heart of the pipeline. It does five things in one call:

1. **Drops a grid** of `xpts × xpts` sample points over the classified image. `xpts = 200` means a 200×200 grid.
2. **Visits each grid cell** and reads its color class.
3. **Builds a transition matrix T** : for each pair of horizontally-adjacent grid cells, increment `T[c1, c2]`. Then for each vertically-adjacent pair, increment `T[c1, c2]` again — but kept as a *separate* horizontal- T and vertical- T .
4. **Excludes background** (if `bkgID + exclude` set): drops the row + column for the white-bg class.

5. Computes the seven Endler metrics from the (filtered) horizontal-T and vertical-T.

The A statistic is just `sum(off_diag(vertical_T)) / sum(off_diag(horizontal_T))`:

- A vertical-bar fish has lots of yellow-to-black horizontal transitions but very few vertical transitions within a bar. Vertical edges (between bars) » horizontal edges within bars. **A > 1**.
- A horizontal-stripe fish: opposite. **A < 1**.
- A reticulate / spotty fish: roughly equal in both directions. **A ≈ 1**.

i Why `xpts = 200`, not 50, not 1000?

- **xpts = 50** → too sparse; the grid skips small features (e.g., the cleaner wrasse stripe is only ~20% of the body, so at 50 pts you sample it at maybe 10 cells, with high variance).
- **xpts = 1000** → expensive (400× more work) and over-samples noise; small JPEG artifacts start showing up as transitions.
- **xpts = 200** → empirically robust on Endler’s example datasets and our test fish. This is the default in his 2012 paper.

2.5 Why `coldists` matters for `m_dS` and `m_dL`

Without a `coldists` table, `adjacent()` returns NA for the boundary-strength metrics. The reason: `m_dS` (mean chromatic boundary strength) is the *average chromatic distance across edges, weighted by edge frequency*. To compute that you need to know the chromatic distance between every pair of color classes — that’s `coldists`.

In `walkthrough.R` we compute a *photographic* `coldists` from RGB (Euclidean distance in RGB space). This is a stand-in for *perceptual* `coldists`, which would come from `vismodel()` %>% `coldist()` once we have a fish-eye visual model in Lab 6.

So: the `m_dS` and `m_dL` students will see today are “geometric saturation steps in RGB,” not “JNDs as a fish would see them.” Be ready for the question; flag the upgrade in Lab 6.

2.6 The three demo fish — what to expect

Demo fish	Family	Pattern	Predicted A	Measured A (typical)
Sergeant major	Pomacentridae	Vertical bars	A > 1	~ 1.5
Cleaner wrasse	Labridae	Horizontal stripe	A < 1	~ 0.6
Mandarin fish	Callionymidae	Reticulate	A ≈ 1	~ 1.0

Tiny variation between runs (± 0.05) is normal — k-means seeds and slight differences in `xpts` grid alignment.

Part 3 — Step-by-step walkthrough of scripts/walkthrough.R

This is the section to keep open while you teach. Each subsection is one numbered step in the script.

Step 0 — Setup

```
library(pavo)           # loads getimg(), classify(), adjacent()
library(magick)          # image I/O backend pavo uses
library(tidyverse)       # bind_rows, pivot_longer, ggplot
source("scripts/helpers.R") # ← critical: loads simple_coldists,
                           #   pick_white_bg, pick_col, segment_fish,
                           #   composite_to_white, flip_to_left_lateral
packageVersion("pavo")
```

What’s happening: loading the three packages plus the helpers, plus checking we’re on pavo 2.9. Each line carries different functions — see the troubleshooting table below for which “could not find function” error means which line was skipped.

Why this matters: if any of these errors out, nothing downstream works. Get students to “all four loaded clean” before proceeding.

The two pedagogical moments to emphasize:

1. **The `source("scripts/helpers.R")` line.** Students forget this when they jump into Part B from a fresh session. Most “could not find function ‘simple_coldists’” errors trace back to it.
2. **The “after Restart R” rule.** Restarting R (Session → Restart R, container restart, Binder reload) wipes the entire environment. Students often try to recover by re-running just one or two lines of Step 0 — but they need ALL four. The `library(pavo)` line in particular is the one they re-skip because the immediate previous error was about a helper function, so they assume the problem is helpers-only.

Troubleshooting Step 0

Symptom	Cause	Fix
there is no package called 'pavo'	install never ran or hit an error	<code>install.packages("pavo")</code> and watch for non-zero exit status; check R --version 4.3

unable to load shared object ... magick.so	system ImageMagick missing	macOS: brew install imagemagick; Linux: sudo apt install libmagick++-dev. Then <i>reinstall</i> the R magick package so it relinks
package 'pavo' was built under R version 4.X warning	pavo binary built on newer R than running	warning only, ignore — works fine

Step 1 — Load the three demo fish (segmented)

```
sergeant_major <- getimg("images/demo/segmented/sergeant-major.jpg")
cleaner_wrasse <- getimg("images/demo/segmented/cleaner-wrasse.jpg")
mandarin_fish <- getimg("images/demo/segmented/mandarin-fish.jpg")
```

What’s happening: `getimg()` reads a JPG into pavo’s `ring` class — internally a 3D array (height $\times$ width $\times$ {R, G, B}) with a few attributes (`imgname`, the original filename).

Why three fish, why these three: they were curated specifically to span the A axis cleanly. The sergeant major has near-perfect verticality (bars top-to-bottom of the body); the cleaner wrasse has a dominant horizontal black stripe with weak vertical structure; the mandarin’s labyrinthine pattern is roughly isotropic in horizontal vs. vertical edges. If a student asks why we don’t pick X — the answer is “the demo is calibrated for clean teaching; in Part B you’ll see messier real-world A values.”

Plot the three side-by-side:

```
par(mfrow = c(1, 3))
plot(sergeant_major); title("sergeant major")
plot(cleaner_wrasse); title("cleaner wrasse")
plot(mandarin_fish); title("mandarin fish")
par(mfrow = c(1, 1))
```

Pause here. **Have students predict A** before computing anything. This is the pedagogical hook — they need to commit to a prediction before seeing the answer.

 Calling on students at this point

Pick any student and ask: “Which one will have the highest A? Why?” If they say “the bars one because vertical edges,” you have a class that’s tracking. If they hedge, push: “What does A measure?” Get them to surface the definition before you reveal numbers.

⚠ Troubleshooting Step 1

Symptom	Cause	Fix
cannot open the connection / cannot open file	working directory is wrong, OR the file is in the wrong folder	First check <code>getwd()</code> . Docker/Binder: should be <code>/home/rstudio/lab</code> . Native: the local repo path. Then <code>list.files("images/student-images")</code> and <code>list.files("images/student-fish/<lineage>")</code> — confirm their file is in one of them. Most common cause: student dropped the JPG into <code>~/Downloads</code> and tried to load with a relative path. Move it to <code>images/student-images/</code> .
Image loads but <code>plot()</code> shows it black or distorted	rare — corrupted JPG	re-clone the repo; check the file is non-empty (<code>file.size("images/demo/segmented/sergeant-major.jpg")</code>)
Error: could not find function "getimg" (or <code>classify</code> , <code>adjacent</code>)	<code>library(pavo)</code> skipped (often after Session → Restart R)	back to Step 0 — re-run all four lines
Error: could not find function <code>"simple_coldists"</code> (or <code>pick_white_bg</code> , etc.)	<code>source("scripts/helpers.R")</code> skipped	back to Step 0 — re-run all four lines. This is the most common Step 3 error in our test runs
Error: could not find function <code>"bind_rows"</code> (or <code>pivot_longer</code> , <code>ggplot</code>)	<code>library(tidyverse)</code> skipped	back to Step 0 — re-run all four lines

Step 1.5 — Segmentation (the conceptual punchline)

This is *the* step to spend time on. Most lab questions trace back to a misunderstanding here.

The visual demo: open three images side-by-side from `images/demo/`:

- `images/demo/raw/sergeant-major.jpg` — what came off Wikipedia
- `images/demo/sergeant-major.jpg` (if present — cropped to bounding box)
- `images/demo/segmented/sergeant-major.jpg` — what we use

Walk through visually: each step strips more of “the photograph” away from “the fish.” The metrics get cleaner as the photo gets cleaner.

Conceptual explanation to give students (use your own words; this is the gist):

If we ran `classify()` on the raw photo, k-means would happily allocate one or two of our k classes to the blue water and the rocks. We'd lose those classes for the actual fish patterning. And when `adjacent()` drops its grid, half of the cells would land in featureless background and just dilute the directional signal. **Segmentation isn't a cosmetic clean-up — it's the difference between measuring the fish and measuring the photograph.**

Three paths students can use (worksheet has all three; we use Path 1 for the demo):

1. **`segment_fish()`** in R — wraps `rembg` + ImageMagick. Best quality.
2. **Web tool + `composite_to_white()`** — `remove.bg` → transparent PNG → magick composites onto white in R.
3. **Manual crop** in Preview.app — fastest fallback, slightly weaker metrics.

⚠ Troubleshooting Step 1.5

Symptom	Cause	Fix
<code>segment_fish()</code> errors with “rembg failed”	<code>uvx</code> and <code>rembg</code> neither on PATH	confirm <code>uvx --help</code> works in terminal; if yes, in R run <code>Sys.getenv("PATH")</code> — if <code>~/local/bin</code> missing, restart RStudio; still failing, fall back to Path 2 (web tool)
magick: not authorized (Linux)	Debian's ImageMagick <code>policy.xml</code> blocks JPG by default	the <code>.binder/Dockerfile</code> patches this; for native Linux, <code>sudo nano /etc/ImageMagick-6/policy.xml</code> and change <code>rights="none"</code> to <code>rights="read write"</code> for JPEG and PNG
Output JPG looks like the fish has been “hollowed out”	<code>rembg</code> 's u2net model failed on this fish (rare for fish but happens for unusual angles)	fall back to Path 3 (manual crop); note the <code>rembg</code> model is trained on humans + objects, not fish-specific
Web tool returns transparent PNG, but <code>getimg()</code> says “image is empty”	student forgot <code>composite_to_white()</code> and tried to load PNG directly	<code>composite_to_white()</code> first, then <code>getimg()</code> on the resulting <code>*-segmented.jpg</code>

Step 2 — Classify each image

```
sm_class <- classify(sergeant_major, kcols = 4)
cw_class <- classify(cleaner_wrasse, kcols = 4)
mf_class <- classify(mandarin_fish, kcols = 6)
```

What’s happening: k-means on RGB. Each pixel snapped to one of k color centroids. Returns an ring of integers (1 to k) with a `classRGB` attribute.

Why these particular k values:

- **Sergeant major ($k=4$):** yellow body + black bar + pale belly + white background = 4
- **Cleaner wrasse ($k=4$):** dark stripe + mid-blue body + light belly + white bg = 4
- **Mandarin ($k=6$):** orange + deep blue + mid blue + dark + accent + white bg = 6 (more colors → more classes)

The +1 rule: segment-to-white means we add 1 to k relative to the number of fish colors, to give the white background its own class.

💡 Live-coding tip — show `classify` on the wrong k

Run `classify(sergeant_major, kcols = 2)` and plot it. Students will see the yellow-and-black bars become **one** class with white as the other — the pattern is destroyed. Then run with $k=4$ and show how the bar structure pops back. This makes the “ k matters” claim visceral.

⚠ Troubleshooting Step 2

Symptom	Cause	Fix
Classified image looks washed out / 2 colors when $k=4$	one class collapsed because its mean was very close to another	re-run; k-means random init can occasionally land on a degenerate minimum
<code>classRGB</code> shows two near-identical rows (e.g., two pale grays)	k is too high for the actual color diversity	reduce k by 1
Error: 'kcols' missing	typo, e.g., <code>kcol = 4</code>	spelled <code>kcols</code> (with the s)

Inspect the palettes: the white-bg class is “the one with R G B 1.” We use this fact in Step 3 to find and exclude it automatically with `pick_white_bg()`.

Step 3 — Adjacency (the heart of the lab)

This is the step that produces the numbers. Students don’t need to follow every line — they need to follow the *concept*.

```
sm_cd <- simple_coldists(sm_class)      # pairwise RGB distances
sm_bkg <- pick_white_bg(sm_class)      # which class is white
sm_adj <- adjacent(sm_class, xpts = 200, xscale = 5,
                  coldists = sm_cd,
                  bkgID = sm_bkg, exclude = "background")
```

What’s happening, in three sentences:

1. `simple_coldists()` computes pairwise RGB Euclidean distances between every pair of color classes. Tells `adjacent()` what counts as a “big” boundary vs. a “small” one.
2. `pick_white_bg()` finds the brightest class (highest R+G+B sum) and returns its index. That’s the white-background class.
3. `adjacent()` drops a 200×200 grid, counts horizontal and vertical color-class transitions, *excludes* the white-bg class from the count, and computes the seven metrics.

The `xscale = 5` argument: ostensibly the body length in cm. *Only matters if you care about distance metrics in real-world units.* For this lab we care about ratios (A) and density (m), so `xscale` could be 1 or 100 with no effect. Tell students “leave it at 5; doesn’t change the lab.”

Why exclude “background”: without this, the long uniform stretches of white pad the transition counts with within-white-bg pairs (which are zero transitions, but the area inflates the denominator of m). A drops; m drops; the contrast between the three fish gets muddier.

💡 If a student asks ‘why not just crop the white off and skip exclude?’

Cropping to the fish’s bounding box leaves white CORNERS still in the image (any non-rectangular fish silhouette has them). The exclude approach handles arbitrary-shaped fish silhouettes correctly. It’s also faster than re-cropping every fish.

⚠ Troubleshooting Step 3

Symptom	Cause	Fix
<code>m_dS</code> and <code>m_dL</code> return NA	<code>coldists</code> not passed	check <code>simple_coldists(class_img)</code> ran and was assigned to <code>*_cd</code>
All metrics look weird, A way off	<code>bkgID</code> set to the <i>wrong</i> class index	check <code>pick_white_bg(class_img)</code> actually points to white. Compare <code>attr(class_img, "classRGB")[bkg_idx,]</code> — should be <code>~(1, 1, 1)</code>
Error: <code>bkgID</code> must be in 1:kcols	<code>bkgID</code> computed for one fish was used on another	re-compute <code>*_bkg</code> from each fish’s own classified image

xpts = 200 runs slowly (~5 sec per fish)	normal on a laptop	mention to class — bigger grids = slower. 200 is the speed/accuracy sweet spot
Result has weird column names	older pavo version uses Asp instead of A	pick_col() already handles this; students see the metric name from the tibble printout

Step 4 — Extract metrics

The `pick_col()` helper grabs the seven metrics across all three fish into a single tibble. **Nothing computational happens here — it's just data wrangling.** Students don't need to follow every line; they need to read the resulting tibble and confirm A goes 1.5 / 0.6 / 1.0.

💡 Show the print-out and pause

```
# A tibble: 3 × 9
  fish      family      k      m      A      Jc      Jt    m_dS    m_dL
  <chr>    <chr>    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 sergeant_major Pomacent...   4 0.42  1.54  0.71 ...    ...    ...
2 cleaner_wrasse Labridae      4 0.31  0.60  0.62 ...    ...    ...
3 mandarin_fish  Callionym...   6 0.55  1.02  0.81 ...    ...    ...
```

Stop and let this sink in. Ask: “Did the prediction hold?” Get a verbal yes from the class before moving on to the visualization.

Step 5 — Predict vs. measure visualization

```
metrics %>%
  pivot_longer(c(m, A, Jc, m_dL), names_to = "metric", values_to = "value") %>%
  ggplot(aes(x = fish, y = value, fill = family)) + ...
```

What's happening: four metric values × three fish = a 4-panel grouped bar plot. Visual confirmation of what the table already showed.

Why this matters pedagogically: numbers tell, plots sell. The bar chart with A on its own panel is the moment students see the pattern claim made visible. The three families each get a distinct color → easy comparison.

⚠ Troubleshooting Step 5

Symptom	Cause	Fix
Plot is empty / no bars	<code>pivot_longer</code> operated on wrong column names	check <code>colnames(metrics)</code> — <code>m</code> , <code>A</code> , <code>Jc</code> , <code>m_dL</code> must all be present
Plot displays but axes are weird	NA values in some metrics (e.g., <code>m_dS</code> missing)	the plot will skip NAs — note in narration if it happens
Error: object 'pivot_longer' not found	<code>library(tidyverse)</code> skipped	back to Step 0

Step 6 — “Now you try” — student practice

This is where students take over and you become the help desk. **The Step 6 template in `walkthrough.R` now points at a real pre-loaded path** (`images/student-fish/chaetodontidae/chaetodon-auriga.jpg`) so students can run the template as-is to verify their pipeline works, then swap in their assigned-lineage species.

! Two image folders — know which is which when triaging

The repo has TWO Part-B image folders, and you’ll get questions about both:

- `images/student-fish/<lineage>/` — pre-loaded starter species (5 per lineage). Read-only by convention; if a student is dropping their own files here, redirect them to `student-images/`.
- `images/student-images/` — dedicated folder for student uploads. This is where their custom fish should go.

When a student says “*my fish isn’t showing up*,” first ask **which folder did you put it in?** Most often the file is in `~/Downloads` (Native), or floating at the project root, or in the wrong lineage subfolder. Direct them to move it to `images/student-images/`.

The most common student errors during Step 6 are:

⚠ Top student questions during Step 6 (in order of frequency)

1. “**My A came back as 0.4 but I thought it would be 1.2!**” — Almost always either a wrong `k` value or wrong `bkgID`. Walk over, ask `attr(my_class, "classRGB")` — does class 1 look like white? If not, `pick_white_bg(my_class)` to find which class IS white.
2. “**How do I write the path to my fish?**” — Three quick wins to teach them:
 - In Files pane, click the file then **More → Copy Path...** (RStudio puts the relative path on clipboard)
 - In Console, type `i` then press **Tab** — RStudio offers folder/file completions
 - `list.files("images/student-images")` — prints filenames they can copy

3. **“segment_fish() is hanging”** — rembg’s u2net model is being downloaded for the first time (~150 MB). Tell them to wait 60 seconds. If it still hangs, fall back to Path 2 (web tool).
4. **“My fish image is too small / pixelated”** — the xpts=200 grid is too dense for tiny images. Either find a larger source image, or the metrics are unreliable (note this in worksheet).
5. **“I picked three vertical-bar fish — what now?”** — they need diverse pattern types. Send them back to FishView / their lineage folder to find one of each pattern.
6. **“getimg() says cannot open file”** — file path or working directory. Have them run `getwd()` (should print `/home/rstudio/lab` for Docker/Binder, or the local repo path for Native), then `list.files("images/student-images")` to confirm the file is where they think.

Part 4 — Pedagogical playbook

4.1 Time budget for the 90-minute lab

Block	Time	What you’re doing
Demo Part A — walkthrough.R Steps 0–5	35 min	live-code with the three demo fish; pause at each prediction point
Transition + Q	5 min	“Now your turn” — read worksheet Part B Step 1 with the class
Independent Part B	35 min	students run pipeline on their three fish; you triage
Wrap-up + cross-lineage discussion	10 min	pick 2–3 groups to share; compare their A values to the demo
Submit reminder	5 min	due-by-EOD reminder, point to BruinLearn

If you run long on demo, drop the `m_dS` / `m_dL` discussion (they can read it in the worksheet). Don’t drop A — it’s the headline.

4.2 What to emphasize in Part A

The two ideas the lab pivots on:

1. **Segmentation is structural, not cosmetic.** Dwell on Step 1.5. Students who skip this skim of the lab end up with confused metrics in Part B and they don’t understand why.

2. **A is a directional signal.** Spend two minutes on the *intuition* — “vertical edges divided by horizontal edges, weighted by where they appear” — before showing the computed A values.

Everything else is bookkeeping (k choice, white-bg exclusion, simple_coldists).

4.3 What to emphasize in Part B

In their independent practice, push students to:

1. **Predict before computing.** Have them write down their predicted A for each of their three fish *in the worksheet* before running `adjacent()`. Then have them update it after measurement.
2. **Reason about disagreements.** When a fish’s measured A doesn’t match their prediction, that’s the most pedagogically valuable moment of the lab. Don’t let them just write “the A came out at 0.9, weird.” Push for “I predicted $A > 1$ because I saw bars, but the bars are partial / oblique / contiguous with the dorsal stripe, which adds horizontal edges I didn’t count.”
3. **Compare across the room.** In the wrap-up, compare three students from three different lineages. Did the A statistic recover the same pattern dimensions across families that haven’t seen each other in 100M years? (Spoiler: yes, because A is geometric, not phylogenetic.)

4.4 Where students get stuck (and what to do)

Stuck on	What’s actually wrong	Your move
“It’s not running”	almost always a typo or missing <code>library(pavo)</code>	run their session’s <code>search()</code> to confirm <code>pavo</code> is loaded
“k-means feels arbitrary”	they’re right — it is, partially	tell them so. The lab teaches reproducible measurement, not an oracle. Different <i>k</i> gives different metrics; they pick the <i>k</i> that matches the visible color count, then defend their choice
“What does the metric <i>mean</i> ?”	conflating measurement with interpretation	A <i>measures</i> edge-direction ratio. <i>Interpretation</i> — “does this fish look striped to a predator?” — needs Lab 6’s <code>vismodel()</code> . Be explicit about the layering
“My fish doesn’t have an A pattern type”	they picked three reticulate fish from the same family	normal for some lineages (e.g., Acanthuridae has lots of disruptive coloration). Tell them: pick the three that show the most contrast in pattern within their lineage; the A values may all cluster, and that’s a finding worth noting

Part 5 — Worksheet rubric (suggested)

The worksheet has three assessable sections. Suggested scoring out of 10:

Component	Points	What you're looking for
Three completed metric tables (Part B Step 3)	3	tables filled, all 7 metrics present, k chosen reasonably for each fish
Comparison answers — A prediction vs measurement (Part B Step 4a)	2	honest yes/partially/no answer with a <i>reason</i> ; “no, because” is fine if reasoned
Cross-lineage comparison (Part B Step 4b)	2	identifies a similar demo fish, names <i>which metrics</i> are similar/different, not just “they both look striped”
Biological interpretation (Part B Step 5)	3	proposes a plausible selection pressure linked to the <i>measured</i> metrics; doesn't have to be correct, but must be reasoned. “I'm uncertain — here's why” answers full marks if the reasoning is solid

Common reasons to deduct:

- Tables half-empty with no explanation
- “ $A < 1$, $A > 1$, $A = 1$, just like the prediction” with no interpretation when the values disagreed
- Selection hypothesis that doesn't reference any of their measured metrics (“disruptive coloration” with no link to high m or high m_dS)
- Wholesale reliance on the demo fish without any engagement with their own three

What to praise (write inline):

- A student who notices their A values are *less dramatic* than the demo and traces it to the segmentation choice they made
 - A student who picks a “boring” reticulate fish, gets $A = 1$ as predicted, and reasons about why their lineage favors disruptive coloration
 - Anyone who refers back to a lecture concept (Lec 11–12 material on cell palette + pattern mechanism) when interpreting their fish
-

Part 6 — Pre-lab and day-of checklist

Two days before lab

- ☐ Confirm Binder is warm — visit <https://mybinder.org/v2/gh/AlfaroLab/eeb187-pavo-lab/HEAD?urlpath=rstudio> and let it build. Subsequent students hitting the same commit get cached builds.
- ☐ Re-render `lab-05-pavo-intro.qmd` and `pre-lab-installation.qmd` if anything changed. Confirm the PDFs exist.
- ☐ Spot-check one student's installation per launch path (Native, Docker, Binder, Codespaces) is achievable end-to-end.

Day before lab

- ☐ Walk through `scripts/walkthrough.R` top-to-bottom yourself in a fresh RStudio session. Time how long it takes — should be ~25 minutes at a steady pace.
- ☐ Have the three pre-rendered metric tables (sergeant major / cleaner wrasse / mandarin) memorized so you can reveal them on demand.
- ☐ Print or have open this guide's Part 4 (pedagogical playbook).

Morning of lab

- ☐ Restart your laptop's RStudio. Run `library(pavo); library(magick); library(tidyverse); source("scripts/helpers.R")` to confirm everything still loads.
- ☐ Test `segment_fish()` on one of the demo fish to confirm `rembg` + `magick` are still on PATH. If `uvx` was last used > 1 month ago, the model cache may have been GC'd; running it once now downloads it.
- ☐ Take 5 student-fish images (3 you'll demo + 2 you'll reserve for "what if" student questions) and run them through the full pipeline as a smoke test.

In lab (90 min)

- See the time-budget table in §4.1.
- **Don't fall behind.** If demo Part A is ticking past 40 minutes, skip the `m_dS/m_dL` panel of Step 5 and let students read it in the worksheet.
- **Take notes on student questions** that surfaced. We update the troubleshooting boxes in this guide based on what comes up in real labs.

Appendix A — Glossary of metrics

For when a student asks “what does X mean?” in the middle of lab and you want a single sentence:

- **k** — number of color classes the image was reduced to.
- **m** — fraction of grid cells that are at a color boundary, i.e., adjacent to a cell of a *different* class. High m = busy pattern. Low m = uniform pattern.
- **A** — vertical-edge frequency divided by horizontal-edge frequency. **A > 1 = bars; A < 1 = stripes; A = 1 = isotropic.**
- **Jc** — Simpson diversity of *color-class areas*. High Jc = many colors, evenly distributed. Low Jc = one or two colors dominate.
- **Jt** — Simpson diversity of *transitions*. High Jt = many different boundary types. Low Jt = pattern is dominated by one or two boundary types (e.g., black-yellow over and over).
- **m_dS** — average chromatic (saturation/hue) distance across edges. High m_dS = saturated, contrasty edges; low m_dS = subtle ones.
- **m_dL** — same as m_dS but for luminance (brightness). High m_dL = bright/dark contrast at edges.

Appendix B — Image folders

The repo’s `images/` directory has THREE folders, and students will ask about all of them:

Folder	What’s in it	Who puts files there
<code>images/demo/</code>	The 3 demo fish for Part A (sergeant major, cleaner wrasse, mandarin)	Instructor (don’t touch in lab)
<code>images/student-fish/<lineage>/</code>	5 pre-loaded starter species per assigned lineage	Already curated — read-only by convention
<code>images/student-images/</code>	Student uploads (custom fish photos)	Students put their JPGs here

Pre-loaded `student-fish/<lineage>/` species:

Lineage	Species
Chaetodontidae (butterflyfish)	<i>Chaetodon auriga</i> , <i>C. lunulatus</i> , <i>Forcipiger flavissimus</i> , <i>Hemitaenichthys polylepis</i> , <i>Heniochus acuminatus</i>
Pomacanthidae (angelfish)	<i>Centropyge bicolor</i> , <i>Holacanthus ciliaris</i> , <i>Pomacanthus imperator</i> , <i>P. paru</i> , <i>Pygoplites diacanthus</i>
Balistidae (triggerfish)	<i>Balistoides conspicillum</i> , <i>Melichthys indicus</i> , <i>Odonus niger</i> , <i>Pseudobalistes fuscus</i> , <i>Rhinecanthus aculeatus</i>

Lineage	Species
Acanthuridae (surgeonfish)	<i>Acanthurus lineatus</i> , <i>Naso lituratus</i> , <i>Paracanthurus hepatus</i> , <i>Zebrasoma flavescens</i> , <i>Z. xanthurum</i>

Students can use any three of their lineage's pre-loaded species without uploading anything. Or they can bring their own JPG and add it alongside the pre-loaded ones (see worksheet Part B Step 1 for image-folder instructions per OS).

Questions, errors, or improvements: post in the EEB 187 channel or PR against this file.